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PROFESSIONAL EXPERIENCE

2025 – Present, Research Associate, Woods Hole Oceanographic Institution, Woods Hole, MA, Department of Marine Chemistry and Geochemistry

2022 – 2025, Postdoctoral Investigator, Woods Hole Oceanographic Institution, Woods Hole, MA, Department of Marine Chemistry and Geochemistry

EDUCATION

University of Tennessee — Knoxville, US
Ph.D. Environmental Engineering — Aug. 2022

University of Chinese Academy of Sciences — Beijing, China
M.S. Environmental Science — Jul. 2017

University of Copenhagen — Copenhagen, Denmark
M.S. Water and Environment — Mar. 2017

University of Jinan — Jinan, China
B.Eng. Environmental Engineering — Jun. 2014

RESEARCH PINTERESTS

- Environmental health
- Fate and transformation of emerging contaminants
- Waste valorization and circularity
- Greenhouse gas emission
- Sustainable materials and circularity
- Science communication and research translation

PUBLICATIONS

Peer-reviewed Papers:

1. **Sun, Y.**, Wu, X., Zanina, O. G., Rivkina, E. M., Lloyd, K. G., Löffler, F. E., and Vishnivetskaya, T. A. 2026. Incomplete denitrifying bacteria drive N_2O fluxes in ancient Siberian permafrost microcosms. *FEMS Microbiology Ecology*. 102(5), fiag034. https://doi.org/10.1093/femsec/fiag034
2. Liu, S., Li, Y., Zeng, X., **Sun, Y.**, Huang, Z., Du, C., Lang, J., Wang, S., and Jia, Y. 2026. Reversible control of microbial As (III) oxidation by nitrous oxide availability in flooded paddy soils. *Environmental Science & Technology*. In press. https://doi.org/10.1021/acs.est.6c04988
3. Zhou, J., Sang, C., Liu, Z., Li, W., **Sun, Y.**, Zhang, J., Wang, S., and Zeng, X. 2026. Bioconversion of selenite to selenium nanoparticles by *Paenibacillus polymyxa* P10: metabolic basis and potential

for wastewater treatment. *Applied and Environmental Microbiology*. e00704-26. https://doi.org/10.1128/aem.00704-26

4. [#]**Sun, Y.**, James, B. D., Houston, R. K., Edwards, B., Izallalen, M., Mazumde, S., Shankar R., Reddy, C. M., and Ward, C. P. 2025. Temperature controls the lifetimes and microbial communities degrading cellulose diacetate in the coastal ocean. *Environmental Science & Technology*. 59, 36, 19468–19478. https://doi.org/10.1021/acs.est.5c05334
5. James, B. D., **Sun, Y.**, Pate, K., Irving, B., Ward, C. P. 2025. Strategies for designing circular, sustainable, and nonpersistent consumer plastic products: A case study of drinking straws. *Environmental Science & Technology*. 59, 34, 18177–18189. https://doi.org/10.1021/acs.est.5c05448
6. Murdoch, F. K., **Sun, Y.**, Fuller, M. E., Lilly, J., Wilson, J., Mullins, L., Hill, A., Löffler, F. E., and Kucharzyk, H. K. 2025. Presence of micro and nanoplastics affects degradation of chlorinated solvents. *Toxics*, 13(8), 656. https://doi.org/10.3390/toxics13080656
7. Waldrop, M. P., Ernakovich, J. G., Schaefer, S. R., Mackelprang, R., Tas, N., Vishnivetskaya, T. A., Li, Z., Ricketts, M. P., Leewis, M. C., Hewitt, R., Hultman, J., **Sun, Y.**, et al. 2025. Microbial ecology of permafrost soils: populations, processes, and perspectives. *Permafrost and Periglacial Processes*, 0:1–14. https://doi.org/10.1002/ppp.2264
8. [#]**Sun, Y.**, Yin, Y., He, G., Cha, G., Ayala-del-Río, H., Konstantinidis, K. T., and Löffler, F. E. 2024. pH selects for distinct N_2O-reducing microbiomes in tropical soil microcosms. *ISME Communications*, ycae070. https://doi.org/10.1093/ismeco/ycae070
9. James, B. D., **Sun, Y.**, Pate, K., Perri, S. T., Shankar, M., Reddy, C. M., and Ward, C. P. 2024. Foaming enables material-efficient bioplastic products with minimal persistence. *ACS Sustainable Chemistry & Engineering*, 12, 43, 16030–16040. https://doi.org/10.1021/acssuschemeng.4c05822
10. Sun, H., Radosevich, M., **Sun, Y.**, Millet, L., Qian, S., and Zhuang, J. 2024. Pore confinement enhances but surface adhesion reduces bacterial cell-to-cell interaction. *Biology and Fertility of Soils*, 1-10. https://doi.org/10.1007/s00374-024-01841-w
11. Yin, Y., Murdoch F. K., Murdoch R., Yan J., Chen G., Xie Y., **Sun Y.**, and Löffler, F. E. 2024. Nitrous oxide inhibition of methanogenesis represents an underappreciated greenhouse gas emission feedback. *The ISME Journal*, p.wrae027. https://doi.org/10.1093/ismejo/wrae027
12. James, B. D., **Sun, Y.**, Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2024. Strategies to reduce the environmental lifetimes of drinking straws in the coastal ocean. *ACS Sustainable Chemistry & Engineering*, 12, 6, 2404–2411. https://doi.org/10.1021/acssuschemeng.3c07391
13. Yang, S., Shobnam, N., **Sun, Y.**, Löffler, F. E., and Im, J. 2024. The relative contributions of Mn(III) and Mn(IV) in manganese dioxide polymorphs to bisphenol A degradation. *Journal of Hazardous Materials*, 461, 132596. https://doi.org/10.1016/j.jhazmat.2023.132596
14. **Sun, Y.**, Mazzotta, M. G., Miller, C. A., Apprill, A., Izallalen, M., Mazumde, S., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2023. Distinct microbial communities degrade cellulose diacetate bioplastics in the coastal ocean. *Applied and Environmental Microbiology*, e01651-23. https://doi.org/10.1128/aem.01651-23
15. Zhang, L., Yin, Y., **Sun, Y.**, Liang, X., Graham, D., Pierce, E; Löffler, F. E., and Gu, B. 2023. Inhibition of methylmercury and methane formation by nitrous oxide in Arctic tundra soil microcosms. *Environmental Science & Technology*, 57(14), 5655-5665. https://doi.org/10.1021/acs.est.2c09457

16. **Sun, Y.**, Wang, C., May, A. L., Chen, G., Yin, Y., Im, J., Campagna, S., and Löffler, F. E. 2023. Mn(III)-mediated bisphenol A degradation: Mechanisms and products. *Water Research*, 235, 119787. https://doi.org/10.1016/j.watres.2023.119787
17. Xie, Y., Ramirez, D., Chen, G., He, G., **Sun, Y.**, Murdoch, F. K., and Löffler. F. E. 2023. Genome-wide expression analysis unravels fluoroalkane metabolism in *Pseudomonas* sp. strain 273. *Environmental Science & Technology*, 57(42), 15925–15935. https://doi.org/10.1021/acs.est.3c03855
18. Sun, H., **Sun, Y.**, Jin, M., Ripp, S. A., Sayler, G. S., and Zhuang, J. 2022. Domestic plant food loss and waste in the United States: Environmental footprints and mitigation strategies. *Waste Management*, 150, 202-207. https://doi.org/10.1016/j.wasman.2022.07.006
19. **Sun, Y.**, Im, J., Shobnam, N., Fanourakis, S. K., He, L., Anovitz, L. M., Erickson R. P., Sun, H., Zhuang, J., and Löffler, F. E. 2021. Degradation of adsorbed bisphenol A by soluble Mn(III). *Environmental Science & Technology*, 55(19), 13014-13023. https://doi.org/10.1021/acs.est.1c03862
20. Shobnam, N., **Sun, Y.**, Mahmood, M., Löffler, F. E., and Im, J. 2021. Biologically mediated abiotic degradation (BMAD) of bisphenol A by manganese-oxidizing bacteria. *Journal of Hazardous Materials*, 417, 125987. https://doi.org/10.1016/j.jhazmat.2021.125987
21. **Sun, Y.**, Zeng, X., Yang, L., Shi, Y., Chen, X., and Zhuang, J. 2017. Effects of strong reductive process on transformation of heavy metals in protected vegetable soil. *The Journal of Applied Ecology*, 28(11), 3759-3766. https://doi.org/10.13287/j.1001-9332.201711.032

Manuscript in Review/in Preparation:

22. #**Sun, Y.**, Grim, S. L., James, B. D., Reddy, C. M., and Ward, C. P. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. ChemRxiv. In review at *Environmental Science: Processes & Impacts.*
23. Ward, C. P., Schwartz, G., **Sun, Y.**, Messenger, S., Frye-Jones, J., Italiane, D., Swarr, G., Aguilera, M., Rodgers, R. Chronic sunlight-driven mobilization of persistent, bioactive carbon and vanadium from asphalt to downstream aquatic systems. In review at *PNAS.*
24. **Sun, Y.**, Ahern, O., Huber, J., Reddy, C. M., and Ward, C. P. Stable-isotope probing of microbial species driving cellulose diacetate biodegradation in the coastal ocean. In preparation.
25. **Sun, Y.**, Reddy, C. M., and Ward, C. P. A next-generation method for quantifying plastic biodegradation in seawater. In preparation.
26. **Sun, Y.**, Shuai, D., Reddy, C. M., and Ward, C. P. Boosting biodegradation of cellulose diacetate bioplastic by graphitic carbon nitride and sunlight in seawater. In preparation.

Conference Papers:

1. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Is adsorbed bisphenol A (BPA) degradable? *Eleventh International Conference on the Remediation and Management of Contaminated Sediments*. Best paper award.
2. **Sun, Y.**, Im, J., He, L., Erickson R. P., and Löffler, F. E. 2022. Soluble Mn(III)-mediates degradation of adsorbed bisphenol A. *Thirteenth Annual Geosyntec Student Paper Competition*. Third-place award.

RESEARCH GRANT EXPERIENCE

Pending Proposals:

1. Collaborative Research: Assessing the Impacts of Plastic-Derived Organic Carbon on Aquatic Microbiomes Across the Freshwater-to-Ocean Continuum. 2026. **PI.** NSF-CBET. $450,000.
2. From Bioplastic Waste to Value-Added Chemicals: Controlled Biological Carbon Valorization. 2026. **PI.** ACS Early Career Postdoctoral–Faculty Bridge Grant. $125,000. *Independent research funding.*
3. Analysis of Microbial Communities in Enrichment Experiments with Cellulose Esters. 2026. **Co-PI.** Eastman Chemical Company. $45,000.

Awarded Funding:

4. Characterizing the Distribution of Bisphenol A in Pores of Microscale Activated Carbon. Neutron Sciences General User Proposal (IPTS #24752.1), Oak Ridge National Laboratory, 2020. **PI**.

HONORS AND AWARDS

- American Society for Microbiology Future Leaders Mentorship Fellowship, 2024
- Extraordinary Professional Promise Award, University of Tennessee, 2022
- Paper Competition Award, Geosyntec Consultants, Inc., 2022
- Paper Competition Award, Battelle Sediments Conference, 2022
- Graduate Student Senate Academic Support, University of Tennessee, 2022
- Graduate Student Senate Travel Award, University of Tennessee, 2019-2022
- Merit student, University of Chinese Academy of Sciences, 2016
- Outstanding Student Scholarship, University of Jinan, 2010-2013

PARTICIPATION IN EDUCATION PROGRAM

- ASM Future Leaders Mentorship Fellowship Program, 2024-2026
- WHOI Summer Student Fellowship Program, May-Aug 2024, 2025
- Geosciences Education & Mentorship Support Program, 2023-2024
- *Inclusive Teaching Certificate*, University of Tennessee, 2023
- *Course-Based Assessment Certificate*, University of Tennessee, 2023
- Student Mentoring and Research Training Program, University of Tennessee–Oak Ridge Innovation Institute, May-Aug 2022
- Summer School for High School Students, China, Jul-Aug 2014

MENTORING

- Kyle DeCruccio, undergraduate student, Cape Cod Community College, Jun-Aug, 2025
- Mallory Kastner, Ph.D. student, MIT-WHOI Joint Program, Feb-Mar, 2025
- Dominic Italiane, undergraduate student, University of Massachusetts-Boston, 2024-2025
- Shinhyeong Choe, Ph.D. student, Woods Hole Oceanographic Institution, Feb-Mar, 2024
- Kimberly Lutchman, undergraduate student, Queens College, CUNY, Aug 2023-2024
- Devan Linson, undergraduate student, Washington State University, May-Aug, 2022
- Divinity Oliver, undergraduate student, Washington State University, May-Aug 2022
- Jessica Rattray, undergraduate student, Vassar College, May-Aug, 2022

PROFESSIONAL DEVELOPMENT AND SERVICES

Professional Development:

- ACS Postdoc to Faculty Workshop, 2026
- Proposal Writing Workshop, Woods Hole Oceanographic Institution, 2023
- Academic Job Application Workshop, Woods Hole Oceanographic Institution, 2023

Service:

- Poster judge, 2025 AEESP Conference.
- Reviewer for *ACS Environmental Au, Bioresource Technology, Chemosphere, Communications Earth & Environment, Ecotoxicology, Environmental Sciences Europe, Environmental Geochemistry and Health, Environmental Monitoring and Assessment, Environmental Research, Environmental Science & Technology, Frontiers in Microbiology*, *Frontiers in Marine Science, Journal of Geophysical Research – Biogeosciences, Journal of Hazardous Materials Advances*, *Journal of Hazardous Materials Letters, Reviews of Environmental Contamination and Toxicology,* and *Water Research*.

Outreach:

- Massachusetts Region V Science and Engineering Fair, 2024
- Reverse Science Fair, Falmouth High School, MA, 2023

Professional Affiliation:

- American Chemical Society
- American Geophysical Union
- American Society for Microbiology
- Association of Environmental Engineering and Science Professors

SELECTED PRESENTATIONS (presenter underlined)

Oral:

1. **Sun, Y.**, Ahern, O., Huber, J., Reddy, C. M., and Ward, C. P. 2026. Stable-isotope probing of microbial species driving cellulose diacetate biodegradation in the coastal ocean. American Chemical Society Spring Meeting, Atlanta, GA.
2. **Sun, Y.** Environmental pollution and health: insights from plastic degradation. 2025. Morgan State University, Baltimore, MD. ***Invited.***
3. **Sun, Y.** Unravelling plastic degradation in the coastal ocean. 2025. Woods Hole Oceanographic Institution, Woods Hole, MA. ***Invited***.
4. **Sun, Y.**, Frye-Jones, J., Grim, S., James, B., Reddy, C. M., and Ward, C. P. 2025. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. AEESP Conference, Durham, NC.
5. **Sun, Y.**, James, B., Reddy, C. M., and Ward, C. P. 2025. Temperature controls the timelines and microbial communities degrading cellulose diacetate in the coastal ocean. American Chemical Society Spring Meeting, San Diego, CA.

6. **Sun, Y.**, Grim, S., James, B., Reddy, C. M., and Ward, C. P. 2025. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. American Chemical Society Spring Meeting, San Diego, CA.
7. **Sun, Y.** Potential of microbial degradation of plastics in the ocean. 2025. AAAS Annual Meeting, Boston, MA.
8. **Sun, Y.** Understanding the fate of conventional and next-generation plastics in the ocean. 2024. University of Massachusetts, Dartmouth, MA. ***Invited***.
9. **Sun, Y.**, Mazzotta, M. G., Miller, C. A., Apprill, A., Izallalen, M., Mazumde, S., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2024. Unique microbes degrade cellulose diacetate bioplastics in the ocean. American Chemical Society Spring Meeting, New Orleans, LA.
10. **Sun, Y.**, Ward, C. P., James, B., and Reddy, C. M. 2023.Translating fundamental knowledge about the degradation of cellulose diacetate in the environment into high-utility, low-persistence consumer products. Eastman Chemical Company, Kingsport, TN. ***Invited***.
11. **Sun, Y.** 2023. "Bioplastics meet biogeochemical cycles", WHOI's Microplastics Initiative. Woods Hole Oceanographic Institution, Woods Hole, MA.
12. **Sun, Y.**, Mazzotta, M. G., Miller, C. A., Apprill, A., Reddy, C. M., and Ward, C. P. 2023. Distinct microbial communities degrade cellulose diacetate plastics in the coastal ocean. AEESP Conference, Boston, MA.
13. **Sun, Y.** 2022. "Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A", webinar presentation. Geosyntec Consultants, Inc. ***Invited***.
14. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A. Eleventh International Conference on the Remediation and Management of Contaminated Sediments, Nashville, TN. ***Invited***.

Poster:

15. **Sun, Y.**, James, B., Ahern, O., Huber, J., Reddy, C. M. and Ward, C. P. 2026. A systematic evaluation of consumer plastics degradation in coastal environments. Gordon Research Conference, Environmental Sciences: Water. Holderness, NH.
16. **Sun, Y.**, Wu, X., Zanina, O., Rivkina, E., Lloyd, K., Löffler, F. E., and Vishnivetskaya, T. 2022. Nitrous oxide emissions from thawing permafrost: A microcosm study. American Geophysical Union Fall Meeting, Chicago, IL.
17. **Sun, Y.**, Yin, Y., Chen, G., Cha, G., Ayala-del-Río, H., Konstantinidis, K. T., and Löffler, F. E. 2022. pH selects for distinct N_2O-reducing microbiomes in tropical soil microcosms. Eighteenth International Symposium on Microbial Ecology, Lausanne, Switzerland.
18. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A. Eleventh International Conference on the Remediation and Management of Contaminated Sediments, Nashville, TN.
19. **Sun, Y.**, Shobnam, N., Im, J., Chen, G., Yin, Y., Erickson, P. R., and Löffler, F. E. 2019. Degradation of adsorbed BPA by oxidized manganese species. American Geophysical Union Fall Meeting, San Francisco, CA.
20. **Sun, Y.**, Shobnam, N., Im, J., Chen, G., Yin, Y., Campagna, S., and Löffler, F. E. 2019. Biologically mediated abiotic degradation of bisphenol A. American Chemical Society Spring Meeting, Orlando, FL.